Evaluating Physics-Informed Learning for Fluid Dynamics

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Introduction

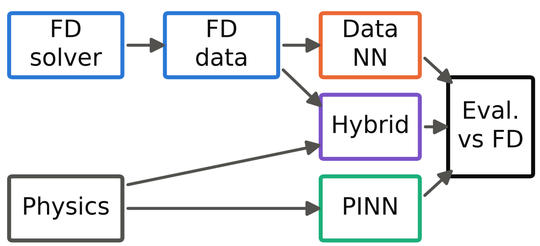
Engineers predict how fluids move (air over a wing, blood in an artery) by solving the equations of fluid motion on a computer. That is accurate but slow. A trained neural network can predict the same flow in milliseconds, but how should it be taught?

I compared three ways of teaching the same network:

- Data NN: learns from the results of finished simulations, so a simulation must run first.
- PINN (physics-informed neural network) [3]: learns only from the laws of fluid motion, with no simulation data.
- Hybrid: learns from both.

RQ1 Which predicts the flow most accurately?
RQ2 Which best respects the laws of physics?
RQ3 Which stays reliable as the flow gets faster?
RQ4 What does each cost to train?

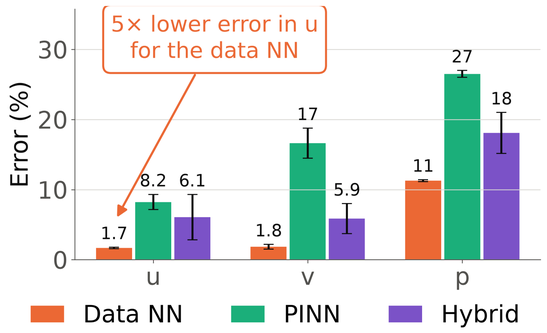
Methods



The test: fluid in a square box, stirred by a lid sliding along the top. A trusted simulation (finite differences, FD), checked against published results [1, 2], is the answer key. All three networks have the same size (17,091 parameters) and the same 20,000 training steps, and are scored on points and moments they never saw. Each is trained 5 times from different random starts to check that results repeat.

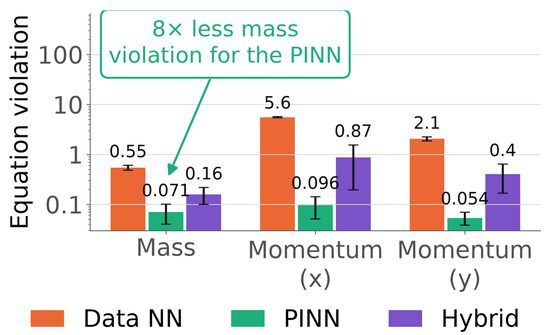
Results

RQ1 accuracy. Error: how far each prediction is from the answer key, in percent, for the speeds u (left-right) and v (up-down) and the pressure p. Lower is better.



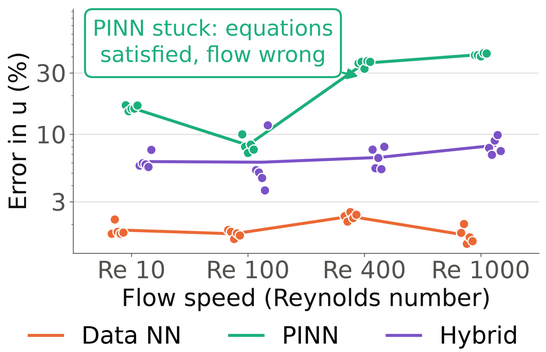
The data NN is the most accurate (1.7% error in u, 5× better than the PINN's 8.2%); the hybrid is in between (6.1%). One hybrid run jumped from 4.1% to 12% at the very end and is reported as it ended.

RQ2 physics. Real fluid cannot appear or vanish (mass) and obeys Newton's laws (momentum). Bars: how much each prediction breaks these rules.



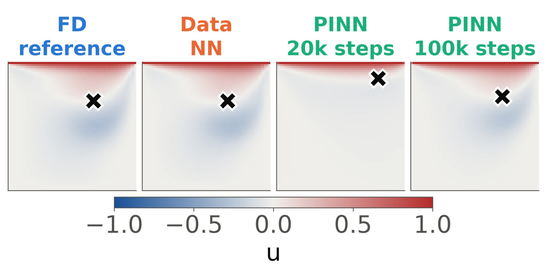
The PINN, which is taught the rules, breaks them 8× less than the data NN (0.071 vs 0.55); the hybrid keeps most of that benefit (0.16). Without its physics lessons (same runs) the hybrid breaks the rules more (0.48); it is then more accurate at Re = 100 (3.0% vs 6.1%) but less accurate at Re = 1000 (9.2% vs 8.2%; in 4 of 5 runs).

RQ3 faster flows. The Reynolds number (Re) measures how fast and swirly a flow is: at Re = 10 it is slow like honey; at Re = 1000 it is fast, with sharp swirls that are harder to learn.



Each dot is one training run; dots close together mean the result repeats. Average error in u at Re = 10, 100, 400, 1000: data NN 1.8, 1.7, 2.3, 1.6%; hybrid 6.1, 6.1, 6.6, 8.2%; PINN 16, 8.2, 36, 42%. Only the PINN gets much worse as the flow speeds up.

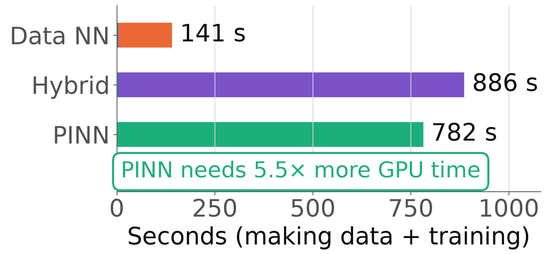
What goes wrong at Re = 400: the PINN puts the center of the swirl (✕) by the lid instead of in the middle, although it obeys the rules better than the data NN (0.03 vs 0.55): obeying the equations is not enough.



Two things help the PINN: training 5× longer (35% → 14%, 3 runs) or starting from the network already trained at Re = 100 (36% → 15%, 5 runs). Adding data, as the hybrid does, gets it right directly: 6.6%. At Re = 1000, 5× longer training still leaves the PINN at 37% (3 runs).

Cost

RQ4. Training time on one graphics card, including the simulation that makes the data.



The simulation took only 3.9 s, so the data NN was 5.5× cheaper than the PINN. Checking the physics at thousands of points every step is what slows the PINN and the hybrid.

Conclusion

	Data NN	PINN	Hybrid
Needs simulation data	Yes	No	Yes
RQ1 Error, Re = 100	1.7%	8.2%	6.1%
RQ2 Mass rule broken	0.55	0.071	0.16
RQ3 Error, Re = 1000	1.6%	42%	8.2%
RQ4 Training time	141 s	782 s	886 s

Green: best of the three. No method wins everywhere. With data, the data NN is the most accurate and cheapest; without data, the PINN is the only option but fails on fast flows. The hybrid is the most balanced: accurate at every speed and close to the physics, at the highest training cost.

References

[1] Marchi, C. H., Suero, R., and Araki, L. K. 2009. The lid-driven square cavity flow: numerical solution with a 1024 x 1024 grid. J. Braz. Soc. Mech. Sci. Eng. 31, 3, 186–198.

[2] Hou, S., Zou, Q., Chen, S., Doolen, G., and Cogley, A. C. 1995. Simulation of cavity flow by the lattice Boltzmann method. J. Comput. Phys. 118, 2, 329–347.

[3] Raissi, M., Perdikaris, P., and Karniadakis, G. E. 2019. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. J. Comput. Phys. 378, 686–707.

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